Fernando Esteban Mejia

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EDUCATION

University of California, Santa Cruz | Santa Cruz, CA

B.A. in Computer Science | GPA: 3.42/4.0 **June 2024**

Relevant Courses: Artificial Intelligence, Computer Graphics, Foundations of Scientific Computing, Data Structures and Algorithms, Computer Systems and C Programming, Database Management Systems, Computational Models, Computer Architecture, Probability Theory, Technical Writing for Computer Science and Engineering

TECHNICAL SKILLS

Programming Languages: JavaScript (Proficient), HTML (Proficient), Python (Proficient), C/C++ (Proficient), SQL (Prior Experience), Fortran (Prior Experience)

Operating Systems: Windows 10/11, Mac, Linux, Unix

APIs & Applications: WebGL (Proficient), Three.JS (Proficient), PgSQL (Prior Experience), VSCode (Proficient), Google Workspace(Proficient), GitHub(Proficient)

WORK EXPERIENCE

Office Assistant | M.A.R.'S Engineering **October 2024 - Present**

- Process and review customer purchase orders, ensuring accuracy and completeness in company systems.
- Prepare detailed production paperwork to facilitate efficient and timely manufacturing.
- Coordinate with stock personnel to verify inventory levels before finalizing paperwork for production.
- Collaborate closely with the president, VP, and production managers to arrange order fulfillment.

PROJECTS

Realistic Dog Park in Three.js | Course Project **April - June 2024**

- Created a realistic 3D dog park using Three.js, featuring multiple dog breeds, various toys, and numerous obstacles, all within a dynamically lit environment.
- Incorporated over 20 primary shapes, utilizing .obj files and comprehensive OBJ loader scripts.

Virtual Maze World in WebGL| Course Project **June 2024**

- Developed an interactive web-based maze game using the WebGL JavaScript API for graphics rendering. Implemented camera controls, camera rotation, in-game character animations, and a real-time FPS counter.
- Ensured cross-browser compatibility and performance optimization.
- Extensively debugged and optimized the game over a two-week period to ensure smooth performance and functionality.

Phong Lighting in 3D World in WebGL| Course Project **May 2024**

- Constructed a virtual 3D world featuring dynamic lighting using WebGL. Implemented Phong shading, interactive camera controls, colored lighting, and various objects to showcase the lighting effects.
- Utilized object-oriented programming to design the scene, organizing multiple objects like cubes and spheres within a hierarchy of classes to ensure precise normal calculations and dynamic lighting.
- Developed HTML buttons and sliders to toggle lighting and normal visualization, providing flexibility in debugging and visual inspection.

Pac-Man AI Search Algorithms | Course Project **April 2024**

- Developed and implemented search algorithms with Python for an AI-controlled Pac-Man to navigate mazes and reach specific goals.
- Applied these algorithms to various maze scenarios, ensuring efficient pathfinding.